

CODEx ROVER BEAN

Robot Body Builder's Guide

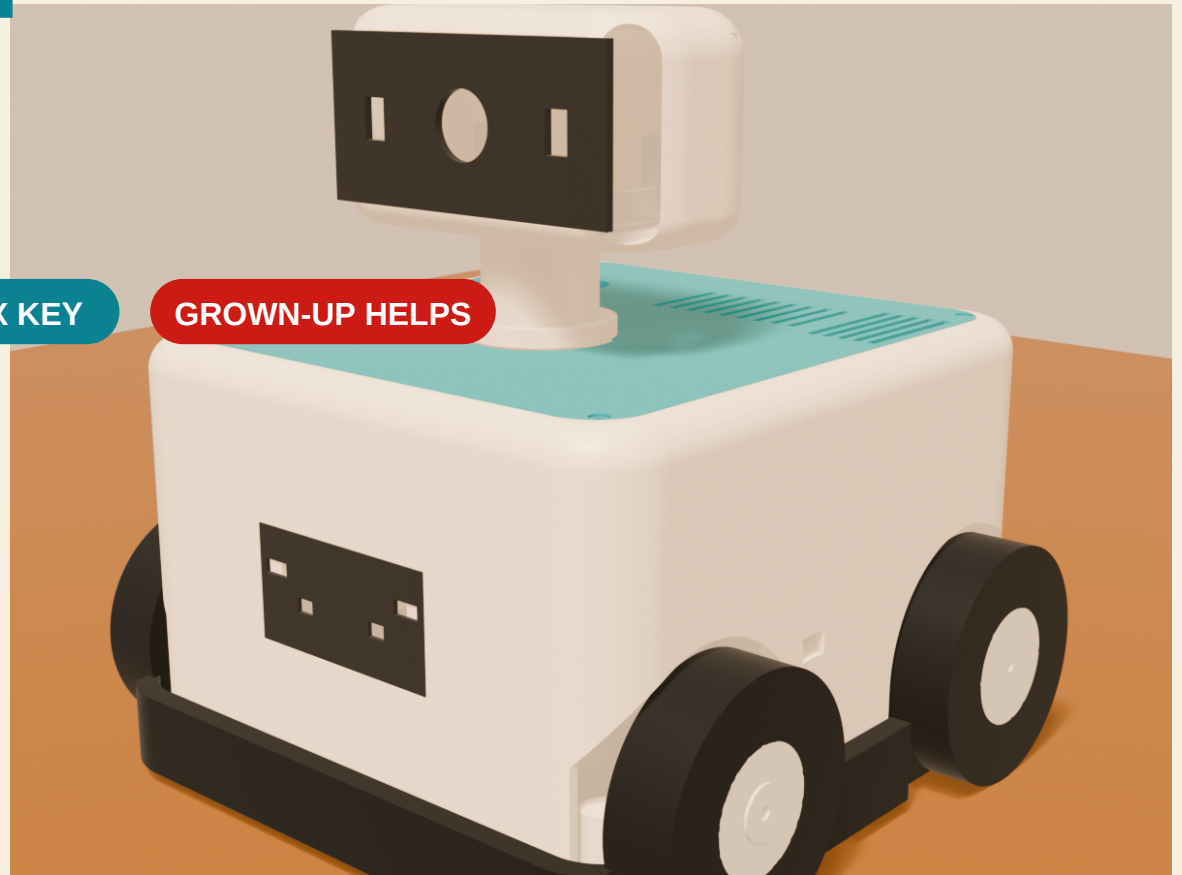
You build it with pictures. One tool. No glue.

AGES 10+

39 PRINTED PARTS

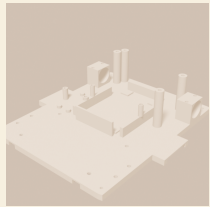
1 HEX KEY

GROWN-UP HELPS



v2 printed-only body - the grown-up pages and the release gates at the back are part of the kit.

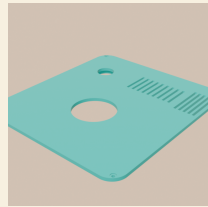
What you printed



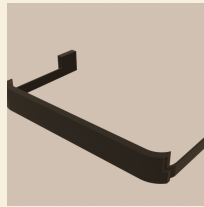
floor tray x1



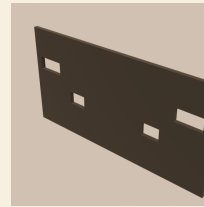
body shell x1



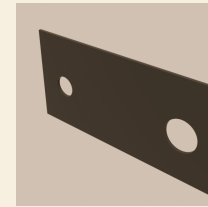
teal lid x1



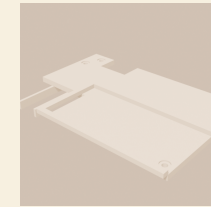
bumpers x2



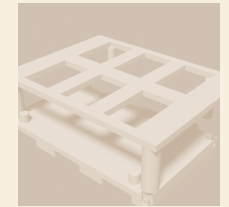
face panel x1



back panel x1



power deck x1



brain tower x1



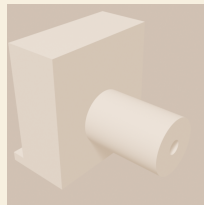
back wheels x2



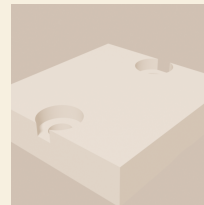
front wheels x2



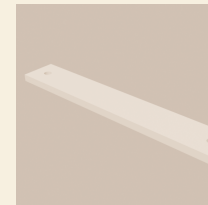
tires x4



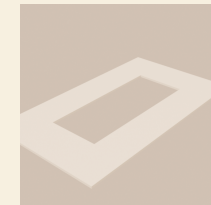
front pods x2



motor caps x2



battery bar x1



battery pad x1



little clamp x1



speaker bars x2



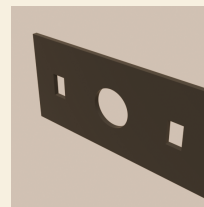
sensor bars x2



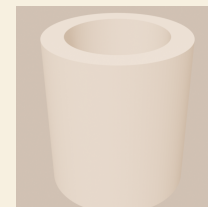
mic ring x1



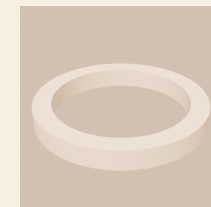
head x1



face x1



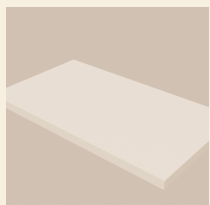
neck x1



neck lock x1



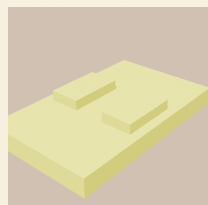
head yoke x1



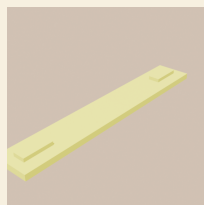
head base x1



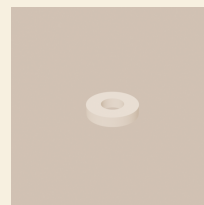
tilt bushing x1



eye glow bar x1



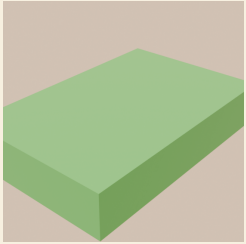
light bar x1



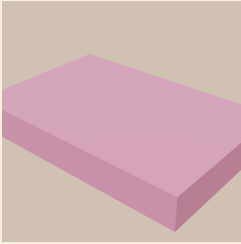
washers x6

The electronics box

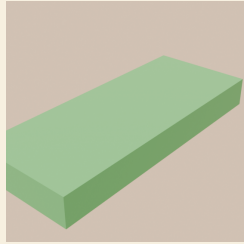
A GROWN-UP HANDLES ALL WIRES



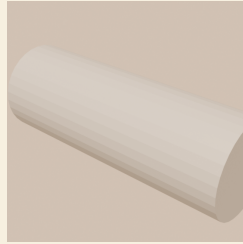
Raspberry Pi 5



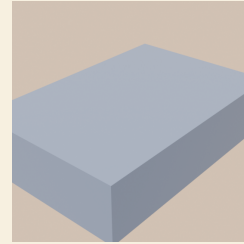
motor board



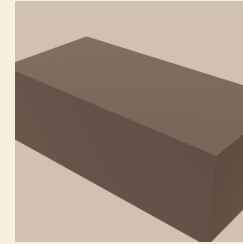
safety board



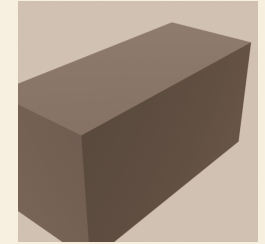
motors x2



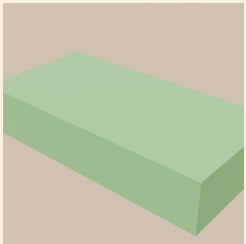
battery



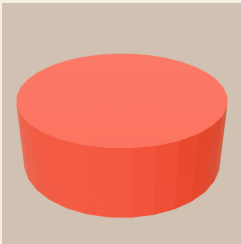
fuse box



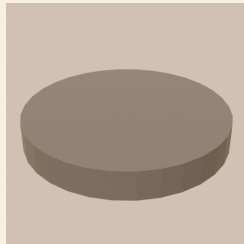
relay



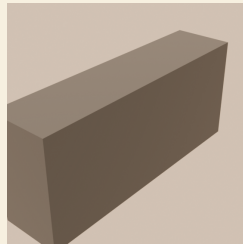
power boards x2



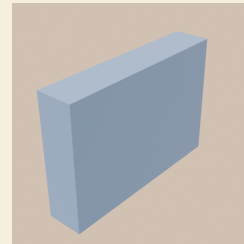
BIG RED BUTTON



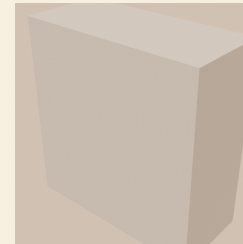
microphone



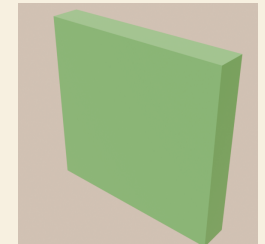
speakers x2



distance eyes x4



neck motors x2



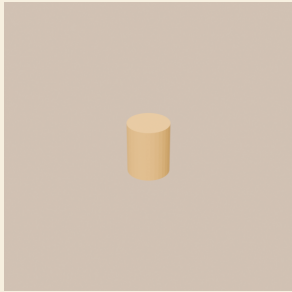
camera

Tools and hardware



M3 x 8 screws x40

Every screw in the robot is this exact screw.



Brass inserts x40

A grown-up melts these in with a soldering iron.

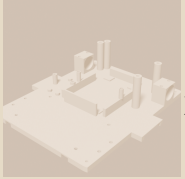
2.5 mm

One 2.5 mm hex key

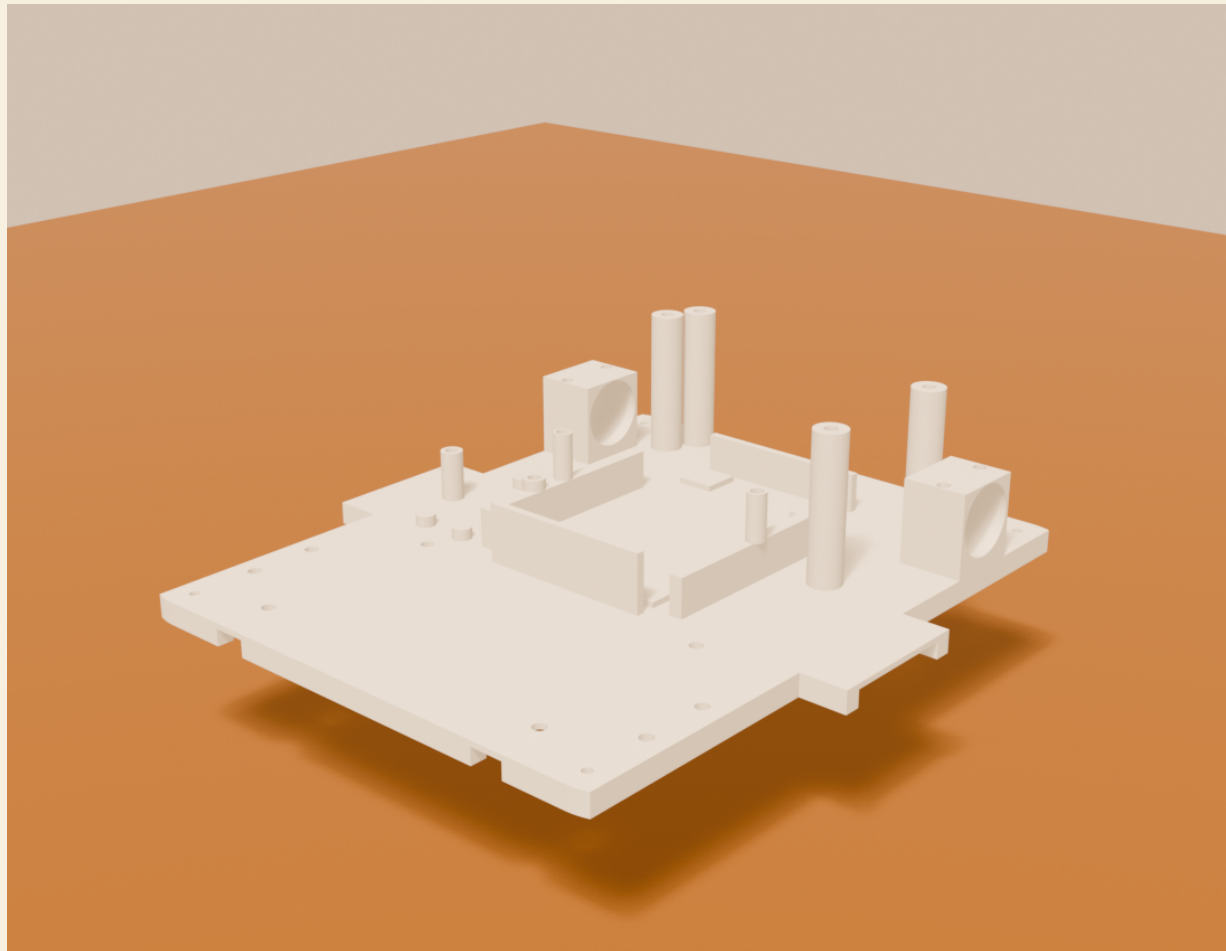
It fits every screw. That's the whole toolbox.

1

Start with the floor



x1



Put the big tray flat on your table. This is the bottom of your robot.

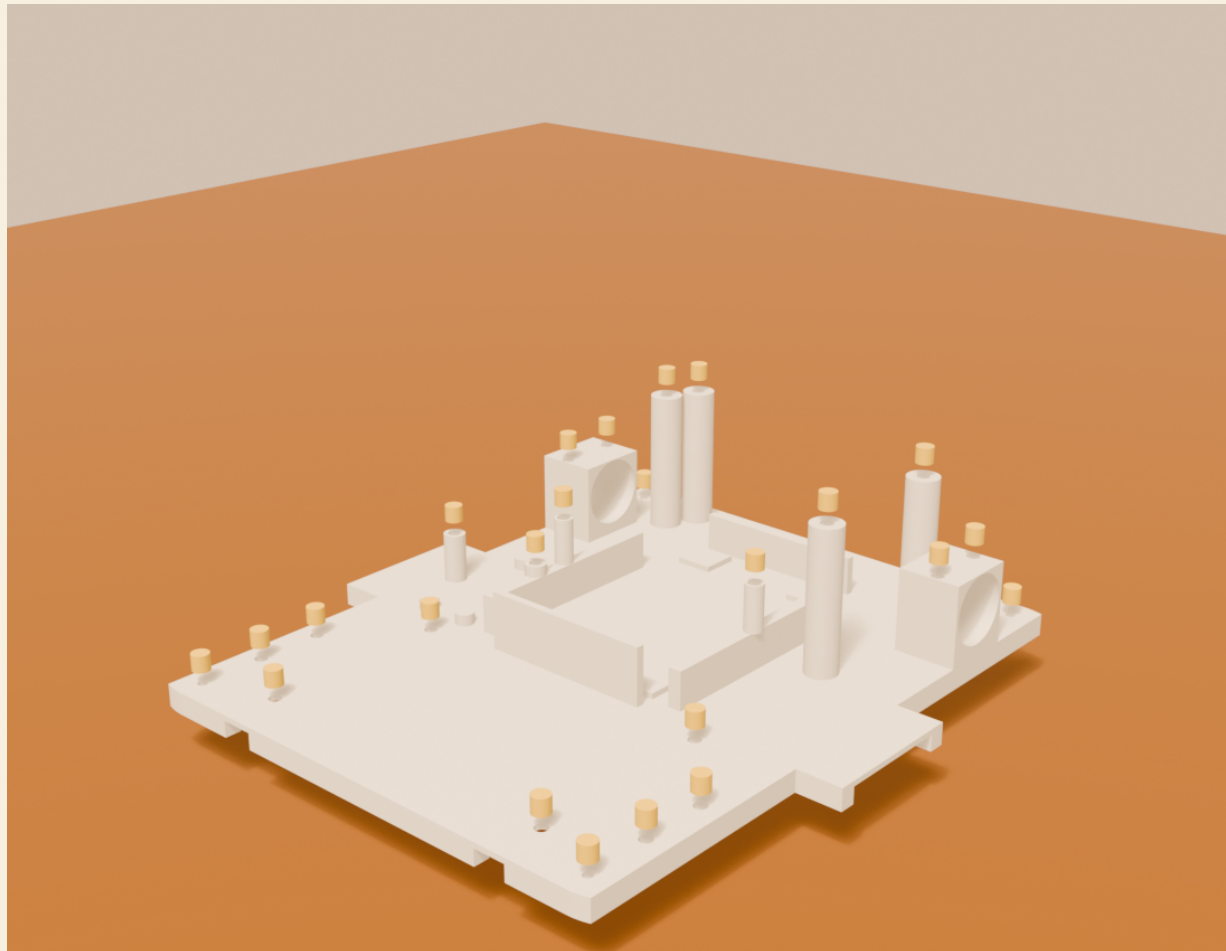
2

Grown-up: melt in the brass inserts

GROWN-UP STEP



x24



A grown-up uses a soldering iron to press a brass insert into every gold spot, flat and straight.

3

Drop in the motors



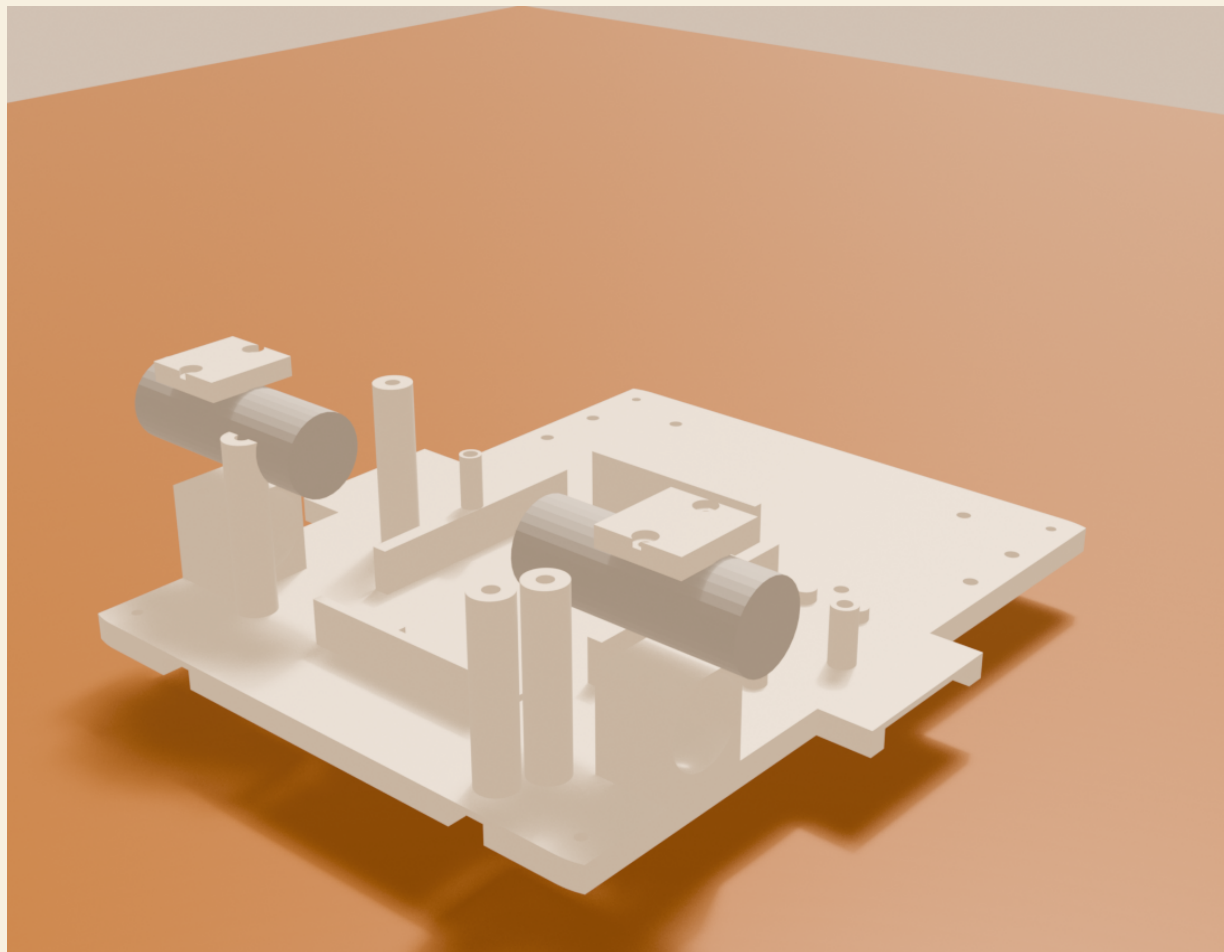
x2



x2



x4
M3 screws



Lay each motor in its cradle, put the little cap on top, and screw it down.

4

Make the wheels



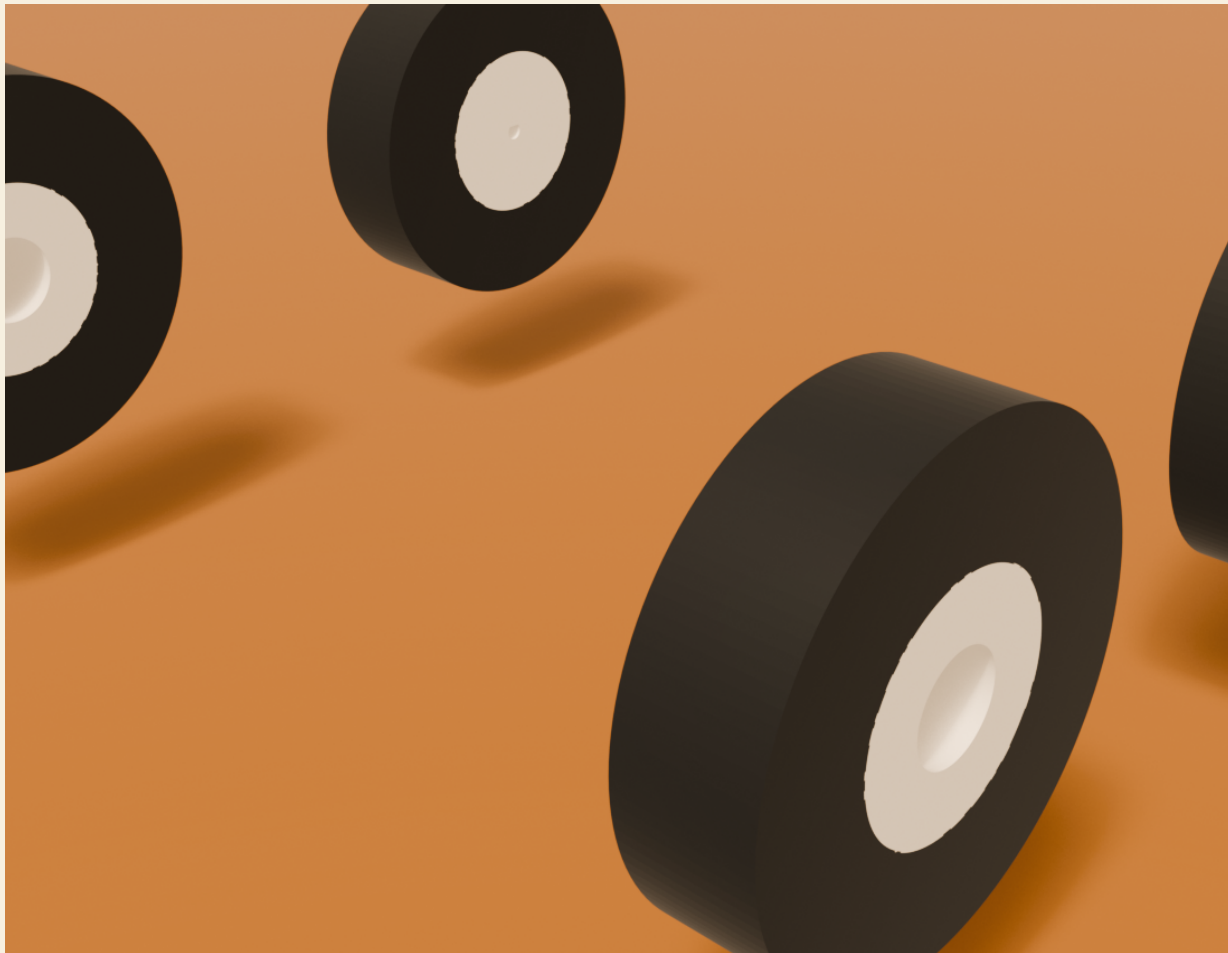
x2



x2



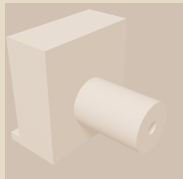
x4



Stretch a rubber tire onto each of the four wheels.

5

Bolt on the front legs

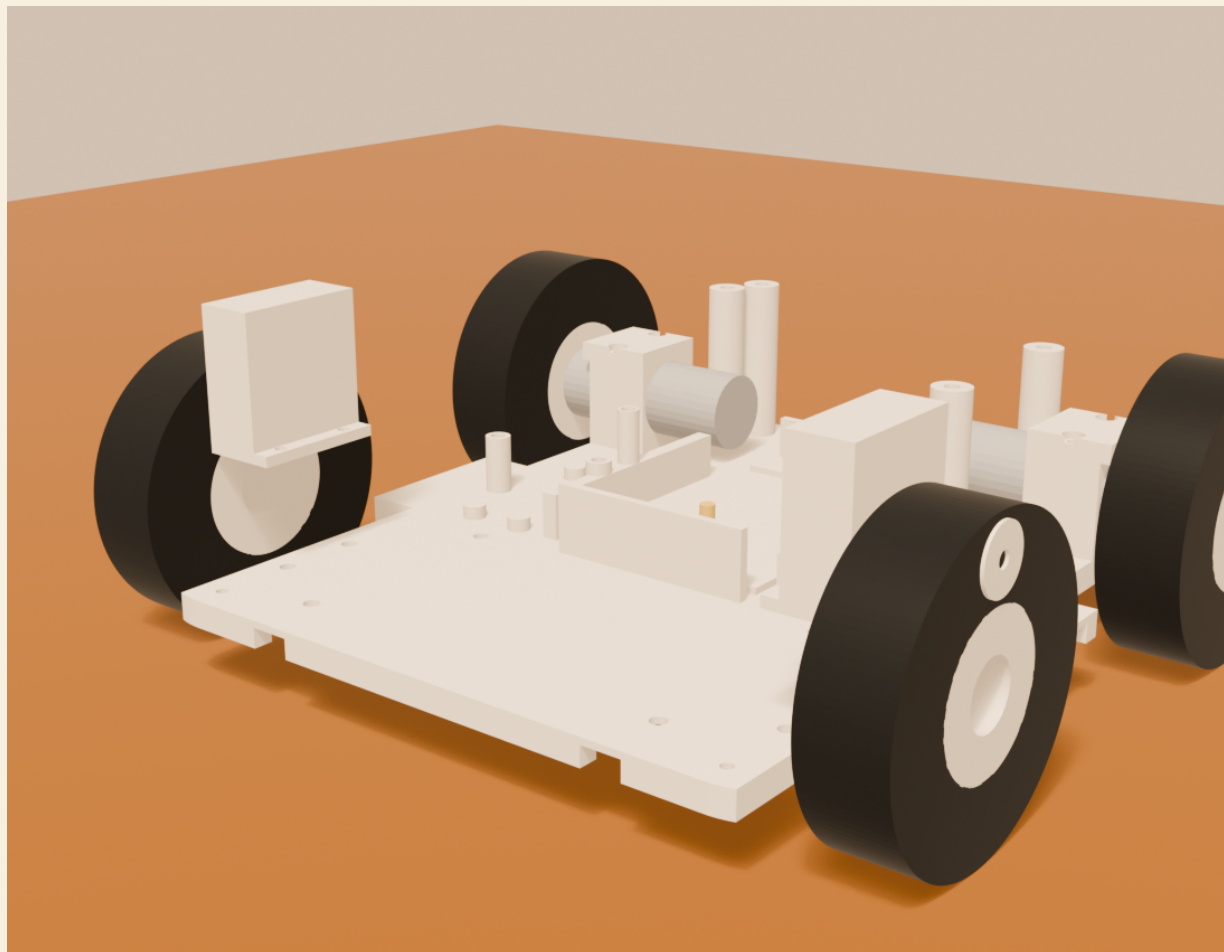


x2



x4

M3 screws



Screw both front pods to the tray. Their round pegs are the front axles.

6

Put the wheels on



x2



x2



x2



x4

M3 screws

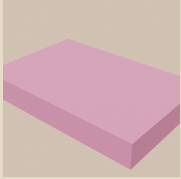
Back wheels push onto the motor shafts, one clamp screw each. Front wheels spin on the pegs: washer first, then screw.

7

Build the brain tower



x1



x1



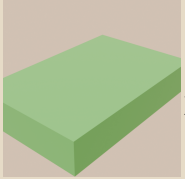
x4

M3 screws

Screw the tower down, then set the purple motor board on its posts.

8

Add the computer

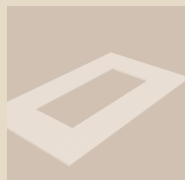


x1

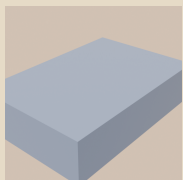
The green Raspberry Pi sits on the tower's top shelf.

9

Strap in the battery



x1



x1



x1



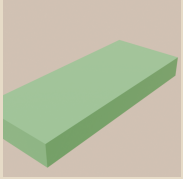
x2

M3 screws

Soft pad down first, battery on top, then the clamp bar holds it tight.

10

Add the safety helper



x1



x1



x2

M3 screws

This tiny green board is the robot's reflexes. Clamp it gently.

11

Grown-up: the power relay

GROWN-UP STEP

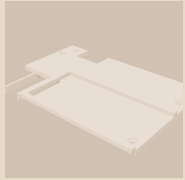


x1

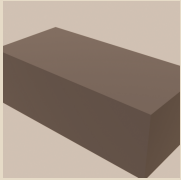
A grown-up mounts the relay. Every wire in this robot is grown-up work.

12

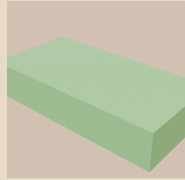
Put on the power deck



x1



x1



x2



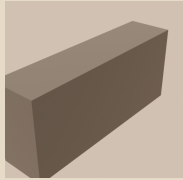
x4

M3 screws

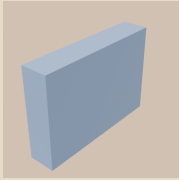
The deck is a shelf: fuse box on top, the two small green boards hang underneath.

13

Speakers and wall-sensing eyes



x2



x2



x2



x2



x6
M3 screws

Speakers rest on their shelves, little blue distance boards behind their windows. Clamp bars hold them.

14

Lower the body shell



x1



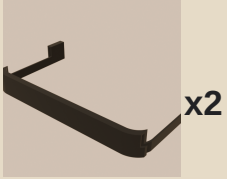
x4

M3 screws

Like a turtle shell! Four screws go up into it from underneath.

15

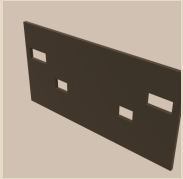
Clip on the bumpers



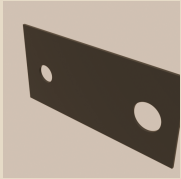
The soft black bumpers wrap the bottom, front and back.

16

Snap in the face and back panels



x1



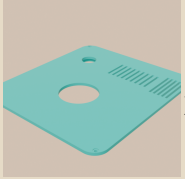
x1

The dark panels click into the front and the back.

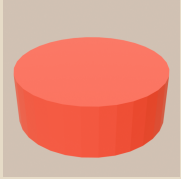
17

Grown-up: the lid and the BIG RED BUTTON

GROWN-UP STEP



x1



x1



x1



x1



x6

M3 screws

A grown-up wires the red emergency stop into the lid, adds the microphone, and screws the lid down.

18

Grow the neck



x1



x1

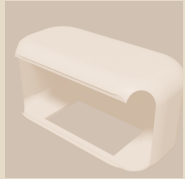


x1

The neck slides through the lid and the collar twists on underneath to lock it.

19

Build the head



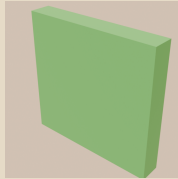
x1



x1



x1



x1



x1

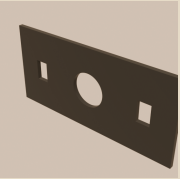


x1
M3 screws

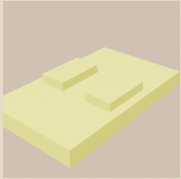
The head shell goes over the yoke; the camera peeks out the front; the little plate closes the bottom.

20

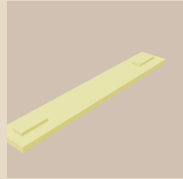
Give it a face



x1



x1



x1

The face panel presses into place, with the glowing eye bar behind it. Say hi!

You built a robot!

High five. Rover Bean's body is done.

For the grown-ups, before ANY power or motion:

- print and pass the 11 calibration coupons first,
- check every purchased part's fit on delivery,
- wire and test the E-stop, bumpers, and watchdog,
- and follow every release gate in the plan docs.

The robot must always fail stopped.

